ThermoQ

User Manual

Version 1.0.2

A worked example on Al–Cu–Li (COST)

Using the calculation files shipped in test/

Switch the interface with Help → Language. Help → User Manual opens this PDF in the language currently selected. Windows users may install ThermoQ-1.0.2-Windows-Setup.exe from GitHub Releases (no Python required).

1. Scope

This manual follows one alloy, Al–Cu–Li, with the COST assessment, from Thermo-Calc and Pandat batch files through extraction and plotting. The figures are computed from the .exp / CSV files in test/; they are the same objects ThermoQ writes to Excel.

Menus: File, Import, Plot, Tools, Help. Calculate sits on the main window (single composition and composition-space batch). After Plot, figures preview in the window (Save As… / Open). Language labels refresh after Help → Language.

Menu	Purpose
Import → Pandat to ThermoQ	Load P.xlsx, Ts.xlsx (required); P-S.xlsx, Ts-S.xlsx (optional).
Tools → Composition Converter	Convert wt% <-> at% compositions.
Tools → Generate Thermo-calc Batch File	Build a .tcm from Template0 + loop body (%Element%, %T%).
Tools → Generate Thermo-Calc Batch Data File	Excel/CSV Data File for Property Model Calculator batch (General Models + Custom/Diffusion).
Tools → Extract Thermo-calc Results	Melting Range, Miscibility Gap, T-zero, TriST Zone (Gibbs; T0 check).
Tools → Generate / Extract Pandat	.pbfx batch (T-zero, Gibbs) and P/Ts, T0.xlsx, TriST Zone.
Plot	Phase surfaces, Qtrue, liquidus vectors, k, T-zero, miscibility gap; in-window preview.
Help → Language	Switch UI English / □□ (menus and open tool windows refresh).
Help → User Manual / Example	Open this PDF (by language) or the bundled Example/ folder.

1.1 Tools → Generate Thermo-Calc Batch Data File

Creates Property Model Calculator batch Data Files (.xlsx / .csv) with English headers (Id, element symbols, Composition unit, Temperature unit; optional Param … / Exp …). Choose a General Model (Coarsening, CET, Crack Susceptibility, Driving Force, Equilibrium, Freeze-in, Interfacial Energy, Liquidus/Solidus, Phase Transition, Scheil, Spinodal, T-Zero, Yield Strength) or Custom/Diffusion (Temperature column, as in Ti–Al–Mn–1173K.xlsx). One Balance (Bal) element plus a composition grid; the temperature grid maps to the model's Param (e.g. Evaluation / Annealing / Start temperature). Phase dropdowns stay on the Thermo-Calc GUI. Sample: Example/Generate Thermo-Calc Batch Data File/.

1.2 Tools → Extract … → TriST Zone (menu settings)

Available under Extract Thermo-calc Results and Extract Pandat Results (Gibbs). TriST region = $\{\Delta G \leq 0\}$ with $\Delta G = Gs - Gl$ (solid-stable / T0-interior). Primary T0 check standard is sheet T0_lines (2D $\Delta G = 0$); T0_tie_1D is only an auxiliary per-temperature summary derived from those lines. TriST_boundaries should match T0_lines (Hausdorff distance and mask□T0 flags appear in Status and TriST_meta). Solid: leave blank for lowest Gs at each (T, C*), or prefer FCC / BCC / HCP. Axes prefer x(*) mole fractions when present; grid N is typically 40–80. Visualize TriST defaults to Overlay T0_lines (check standard). Mask column: below_T0_dG_le_0.

2. Installation

— Windows end users: install ThermoQ-1.0.2-Windows-Setup.exe from GitHub Releases (no Python).

— Developers: Python 3.10–3.12 with tkinter. `pip install -r requirements.txt` then `python main.py`.

— TriST workbooks require matplotlib and scipy. Interactive HTML requires plotly.

3. Case: melting range (Thermo-Calc)

Folder for templates: test/Generate Thermo-calc Batch File-Melting Range/Al-Cu-Li/. Folder for results: test/Extract Thermo-calc Results-Melting Range/Al-Cu-Li_COST/exp/ (filenames such as Al0.050Cu0.010Li_np-T.exp).

3.1 Generate the .tcm

1. Tools → Generate Thermo-calc Batch File.
2. Template0 = template0.txt (database, system, baseline s-c, first make file).
3. Loop template = template.txt. Placeholders %Cu% and %Li% are replaced on the Min/Max/Step grid (mass fraction).
4. Write the merged .tcm (Alltcm.tcm is already in that folder) and run it in Thermo-Calc. Each composition produces one *_np-T.exp.

Template0 (baseline composition, excerpt):

```
go da
sw tcal9
d-sys al cu li
l-s c
get
go p-3
s-c t=500,p=101325,n=1
s-c w(cu)=0.00001
s-c w(li)=0.00001
l-c
c-e
s-a-v 1 t 300 1000 12.5
save_workspace Al0.00Cu0.00Li_np-T.POL y
step normal
post
s-d-a x t
s-d-a y np(liquid)
make file Al0.00Cu0.00Li_np-T.exp
```

Loop body (one block per grid point):

```
back
s-c w(cu)=%Cu%
s-c w(li)=%Li%
l-c
c-e
s-a-v 1 t 300 1000 12.5
save_workspace Al%Cu%Cu%Li%Li_np-T.POL y
step normal
post
s-d-a x t
s-d-a y np(liquid)
make file Al%Cu%Cu%Li%Li_np-T.exp
```

3.2 Extract liquidus and solidus

1. Tools → Extract Thermo-calc Results → Melting Range. Select the exp folder. Default filter *_np-T.exp.
2. The filename supplies w(Cu) and w(Li). Blocks between "$ PLOTTED COLUMNS ARE : T and NP(LIQUID)" and BLOCKEND are read.
3. Liquidus is the lowest T with NP(LIQUID) = 1; solidus is the highest T with NP(LIQUID) = 0. Optional Template1 writes a .tcm for files listed as errors in Status.

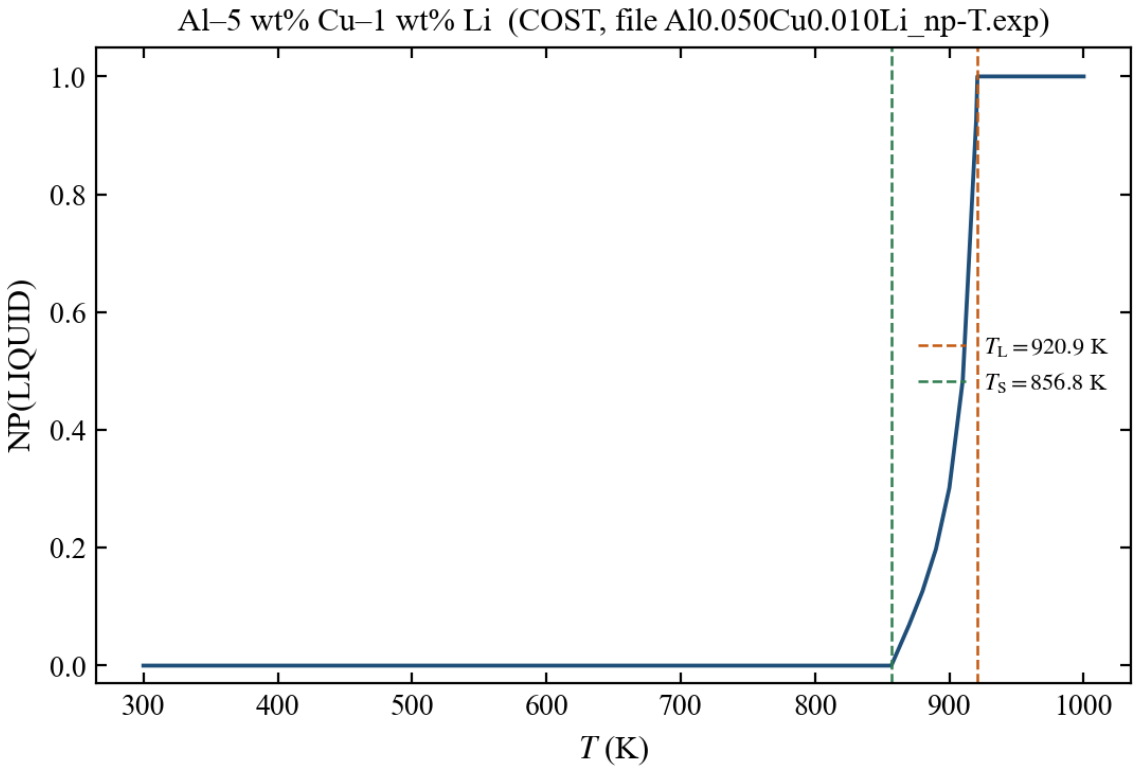


Fig. 1. NP(LIQUID) versus temperature for Al–0.050Cu–0.010Li (COST). Vertical lines are the liquidus and solidus reported in the extract Excel.

Five rows from that extract (same algorithm as the Melting Range tab):

File	w(Cu)	w(Li)	TL / K	TS / K	ΔT / K
Al0.050Cu0.010Li_np-T.exp	0.050	0.010	920.9	856.8	64.1
Al0.100Cu0.020Li_np-T.exp	0.100	0.020	904.2	843.2	61.0
Al0.150Cu0.030Li_np-T.exp	0.150	0.030	880.7	836.8	43.9
Al0.180Cu0.005Li_np-T.exp	0.180	0.005	883.1	814.6	68.5

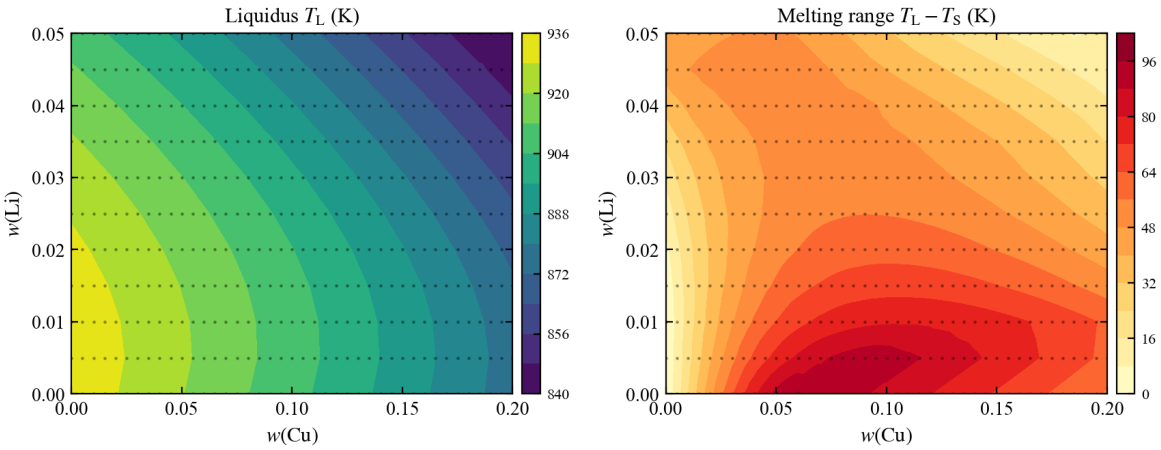


Fig. 2. Liquidus and melting range over the Al–Cu–Li COST grid in test/…/Al-Cu-Li_COST/exp. Dots are individual .exp files. Plot → Phase Surfaces → Thermo-calc loads the Excel of this map.

4. Case: T-zero

Templates: test/Generate Thermo-calc Batch File-T0/Al-Cu-Li/. Results: test/Extract Thermo-calc Results-T0/Al-Cu-Li_COST/exp/AlCu-0.030Li_T0.exp, etc. The loop holds w(Li) fixed and scans w(Cu) from 0 to 0.2.

```
back
s-c w(LI)=%LI%
l-c
c-e
S-A-V 1
w(Cu)
0
0.2
0.005
save AlCu-%LI%Li_T0.POL y
step
t-z
FCC_A1
liquid
po
s-d-a x w(CU)
s-d-a y t-k
make file AlCu-%LI%Li_T0.exp
```

1. Extract Thermo-calc Results → T-zero. Filename → w(Li); XTEXT W(CU) → scan axis; plotted Y → T0 (K).
2. Save t_zero.xlsx. Plot → Plot T-zero Surface (Thermo-Calc), set the isotherm interval.

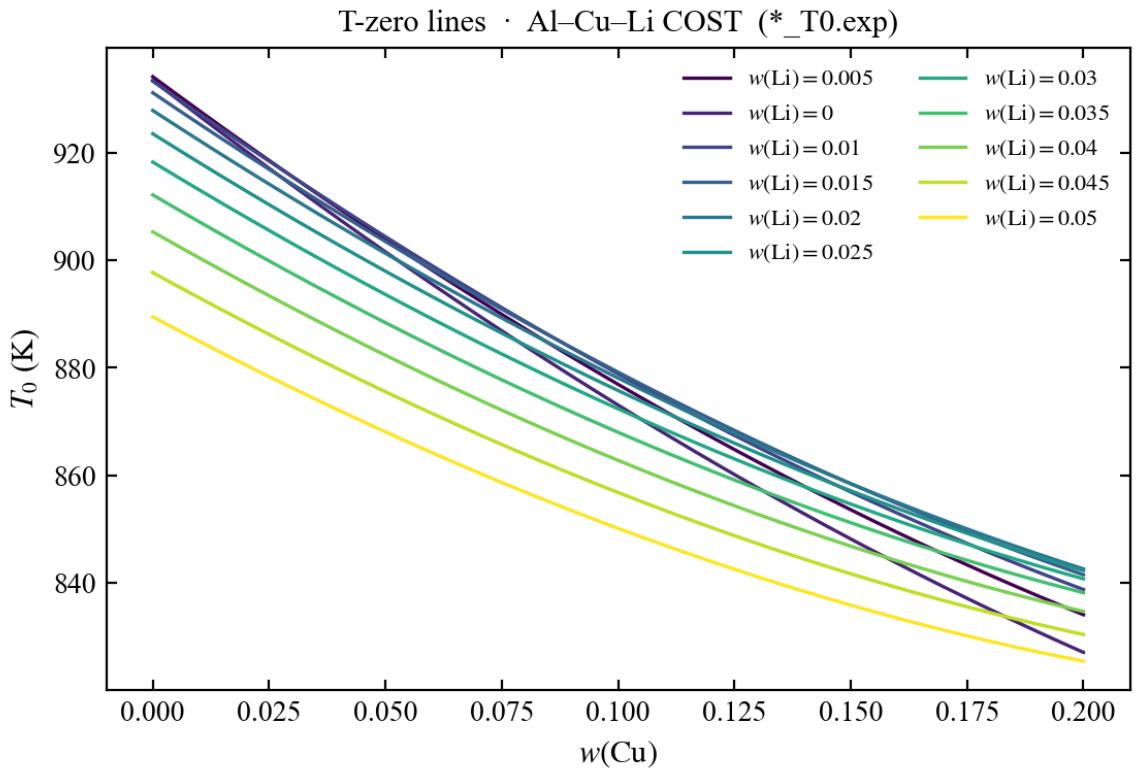


*Fig. 3. T0(w(Cu)) at fixed w(Li) from every *_T0.exp in the COST folder.*

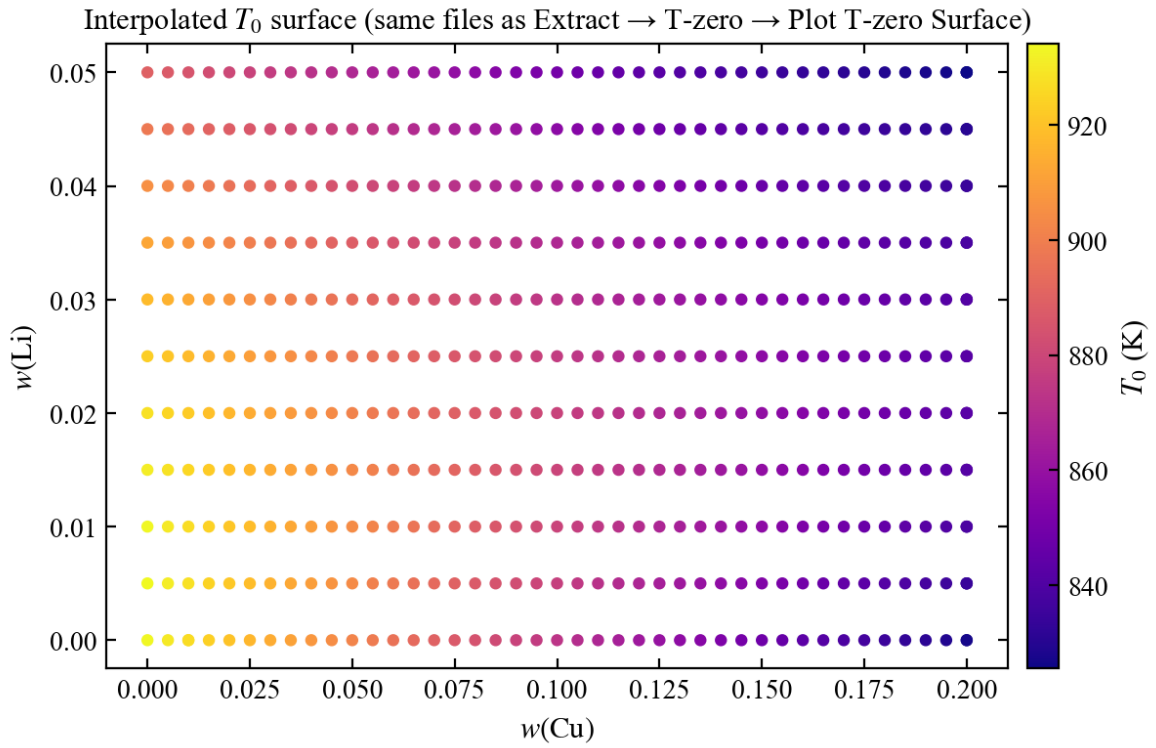


Fig. 4. Linear interpolation of those lines on the w(Cu)–w(Li) plane — the field drawn by Plot T-zero Surface.

5. Case: liquid miscibility gap

Files: test/Extract Thermo-calc Results-Miscibility Gap/Liquid/Al-Cu-Li_COST/exp/AlCuLi_800.exp (one temperature per file). Extract → Miscibility Gap parses T from the name, and XTEXT/YTEXT as mole % Cu and mole % Li.

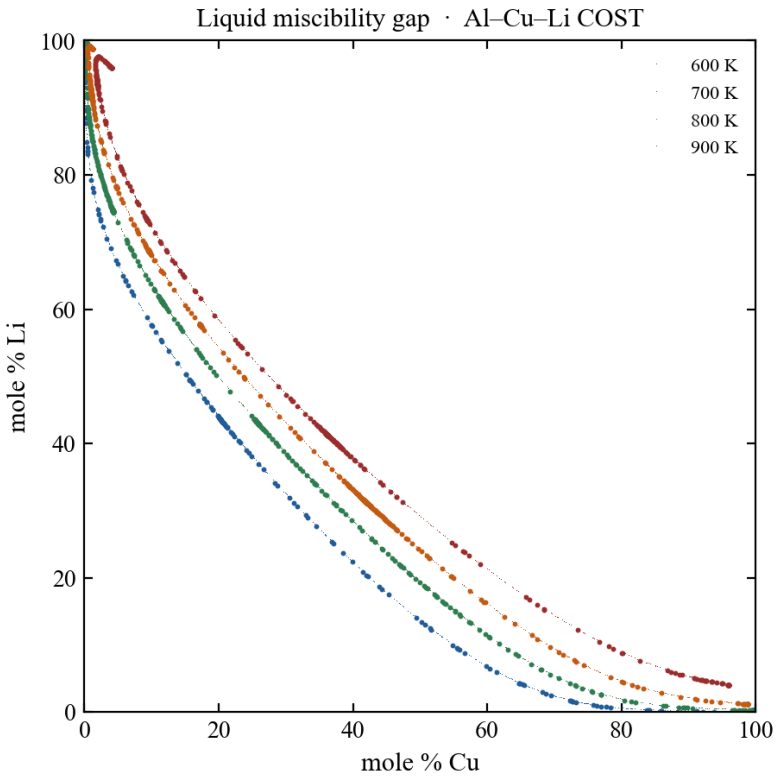


Fig. 5. Liquid miscibility-gap traces at 600–900 K from the COST .exp set. Plot → Plot Miscibility Gap reconstructs the same boundaries (with interpolation at a requested T).

6. Case: Gibbs energies and TriST Zone

Files: test/Extract Thermo-calc Results-Gibbs/Al-Cu-Li_COST/exp/, e.g. AlCu-0.050Li_650T_Gibbs.exp. Each file is one (w(Li), T); the scan axis is w(Cu). Sections are labelled "$ PLOTTED COLUMNS ARE : W(CU) and GMR(PHASE)".

1. Extract Thermo-calc Results → TriST Zone. Folder search is recursive; default filter .*_Gibbs\.exp$. Axes X/Y fill after Browse.
2. Optional Solid preference: blank = lowest Gs; or FCC / BCC / HCP. Build TriST workbook (T0_lines primary; T0_tie_1D aux; TriST_boundaries; TriST_mask with below_T0_dG_le_0; TriST_meta; optional _trist_cube.npz). Grid N is typically 40–80.
3. Check Status / TriST_meta for Hausdorff (T0_lines vs TriST_boundaries) and mask□T0. Failed files go to Status; Template1 can emit one .tcm per error.
4. Visualize TriST (Plotly / 2D / 3D / GIF) with Overlay T0_lines enabled by default.

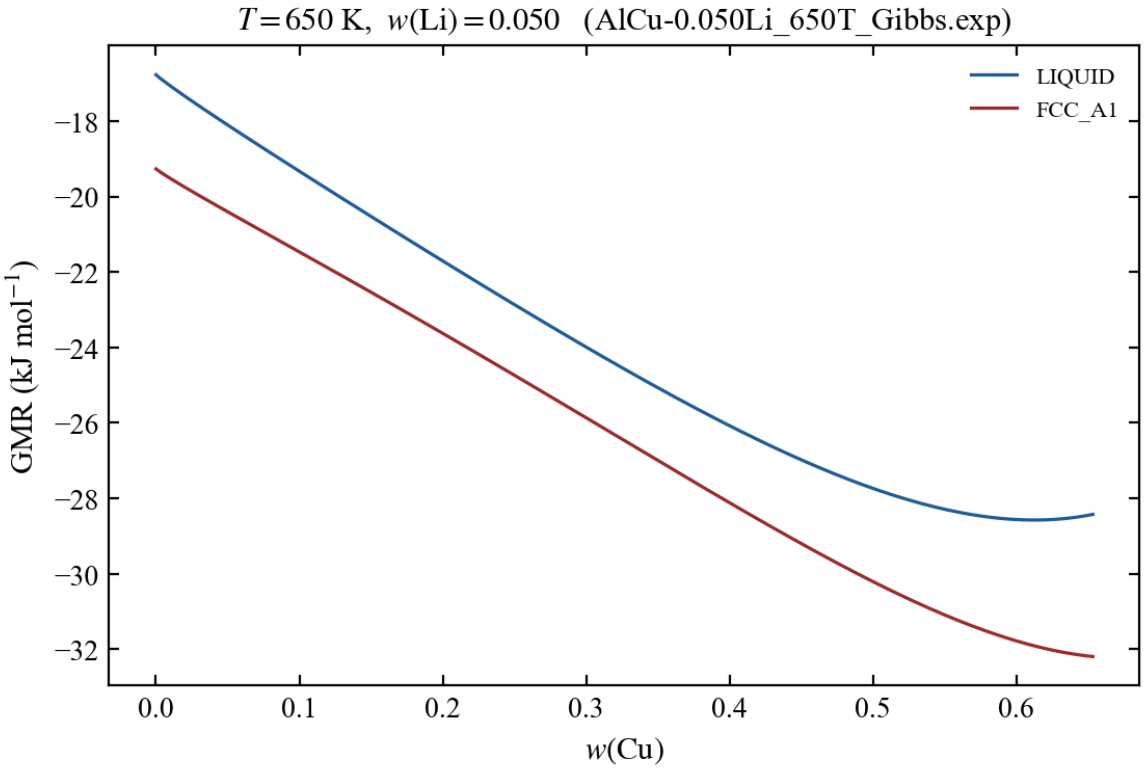


Fig. 6. Molar Gibbs energy of LIQUID and FCC_A1 versus w(Cu) at 650 K and w(Li) = 0.050, taken from one COST Gibbs .exp.

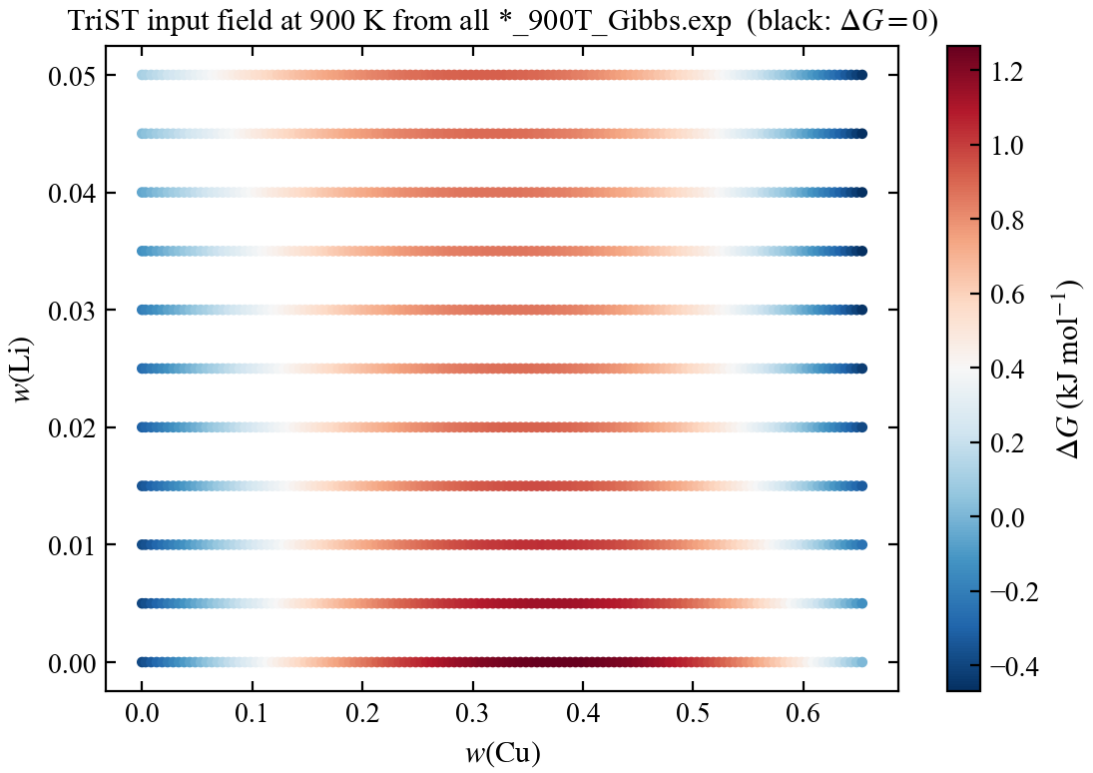


*Fig. 7. ΔG = G(FCC) − G(L) at 900 K assembled from all *_900T_Gibbs.exp in the same folder. The black contour is ΔG = 0 (T0_lines). The TriST region is the solid-stable side ΔG ≤ 0.*

7. Case: Pandat Lever table (same alloy)

Batch templates: test/Generate Pandat Batch File-T0/Al-Cu-Li_COST/ and …-Gibbs/. Solidification tables: test/Extract Pandat Results-P Ts (Lever Scheil)/Al-Cu-Li_COST/. Extract Pandat Results → P/Ts writes P.xlsx / Ts.xlsx / P-S.xlsx / Ts-S.xlsx, then Import to ThermoQ. Calculate uses Newton interpolation on those tables for Qtrue, Q/P/β, ΔT and ΔTs.

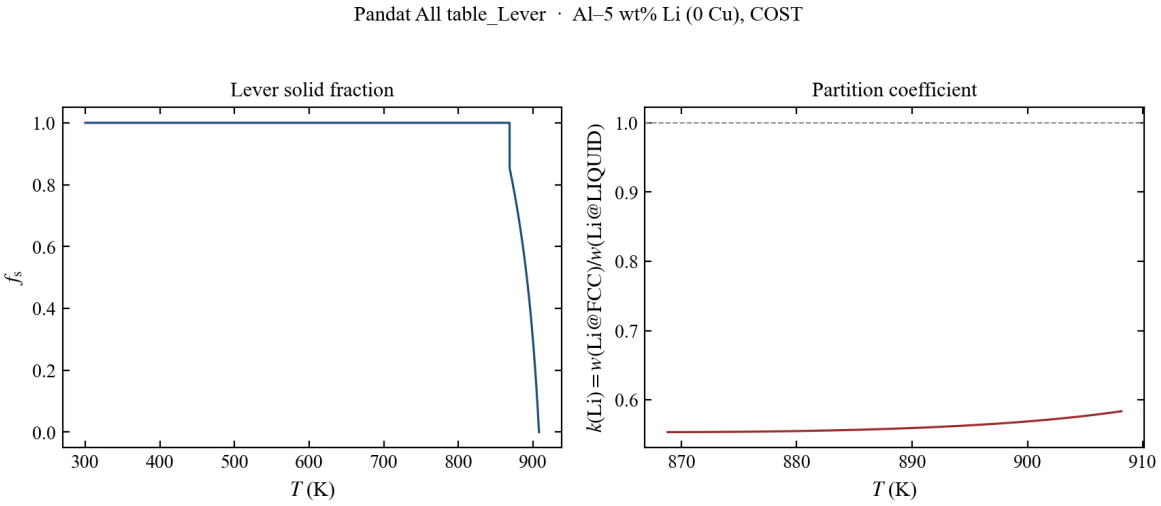


Fig. 8. One Lever CSV from that folder (Al–5 wt% Li, 0 Cu): solid fraction and k(Li). Plot → Solid–Liquid Partition Coefficients uses the same k = w(@solid)/w(*@LIQUID).*

8. Plot tools after extract / import

— Phase Surfaces — Pandat P/Ts or the melting-range Excel of §3.

— Qtrue, liquidus vectors, partition k (liquidus / isotherm / isocomposition).

— T-zero Surface and Miscibility Gap — Excel from §4 and §5.

— Plot Labels: empty fields keep default axis names.

9. Where the other test/ trees fit

test/ folder	ThermoQ command
Generate Thermo-calc Batch File-{Melting Range, T0, Gibbs, Miscibility Gap}/	Generate Thermo-calc Batch File
Extract Thermo-calc Results-Melting Range/	Extract → Melting Range
Extract Thermo-calc Results-T0/	Extract → T-zero
Extract Thermo-calc Results-Gibbs/	Extract → TriST Zone
Extract Thermo-calc Results-Miscibility Gap/	Extract → Miscibility Gap
Generate Pandat Batch File-T0 / -Gibbs/	Generate Pandat Batch File
Extract Pandat Results-P Ts (Lever Scheil)/	Extract Pandat → P/Ts
Extract Pandat Results-T0/	Extract Pandat → T-zero
Extract Pandat Results-Gibbs/	Extract Pandat → TriST Zone

The same layout is repeated for Al–Cu–Si, Al–Mg–Li, Al–Mg–Si and Ti–Fe–Cu, with COST, TCAL9, He2009, Peisheng2011 or Bo2013 in the folder name. Help → Example opens the smaller Example/ tree (includes Generate Thermo-Calc Batch Data File/Ti-Al-Mn-1173K.xlsx).

10. Notes

— Close an extract workbook in Excel before overwriting it.

— Abnormal-point TCM files require a Template1 with %Element% and %T%, after Status has recorded failures.

— TriST needs scipy and matplotlib. T0_lines is the check standard; re-extract after upgrading if you need TriST_meta.